INCH-POUND

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PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY
TYPE 10AKP7

This specification is approved for use by the Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: Magnetic deflection and focus, aluminized screen.

DIMENSIONS AND PIN CONNECTIONS: See figure 1.

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	Ec2	Eb	Ehk	Rg1	Alt
Unit:	V	V dc	V dc	V dc	V dc	Meg	ft
Maximum:	6.9	0 1/	770	11,000	±125	1/	60,000
Minimum:	5.7	-180	----	7,000	---	---	---
Test condition:	6.3	Adjust	650	9,000	---	---	---

See footnotes at end of table I.

GENERAL:

Qualification - Required.

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TABLE I. Testing and inspection.

Inspection	Method	Notes	Conditions	Symbol	Limits		Unit
					Min	Max	
Qualification inspection							
Base material insulating quality	1216	---	Zone 5 (minimum)	---	---	---	---
Electrode current (grid No. 2)	5201	---	Ec1 = 0	Ic2	---	15	µA dc
Cathode illumination	5216	---		---	---	---	---
Direct-interelectrode capacitance	1331	---	Ck to all Cg1 to all Cg2 to all	Ck Cg1 Cg2	--- --- ---	8.0 10.0 10.0	pF pF pF
Pressure (implosion)	1141	---		---	---	---	---
Vibration	5111	---		Width	---	2.0	mm
Barometric pressure, reduced	1002	3/	54 mmHg; maximum voltage applied	---	---	---	---
Conformance inspection, part 1							
Voltage breakdown	5201	---		---	---	---	---
Voltage breakdown (magnetic types)	5201	---		---	---	---	---
Gas ratio	5206	6/		Gr	---	0.25	---
Neck straightness	5101	4/		---	---	---	---
Bulb, screen, and faceplate quality	5106	---		---	---	---	---
Modulation	5223	---	Ib = 200 µA dc	ΔEc1	---	38	V dc
Spot position (magnetic deflection)	5231	---		---	---	18	mm
Zero-bias anode current	5236	---		---	---	---	---
Grid-cutoff voltage	5241	---		Ec1	-40	-70	V dc
Grid No. 1 leakage current	5251	---		---	---	---	---
Aperture alignment	---	2/		Distance	---	8.0	mm

See footnotes at end of table.

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TABLE I. Testing and inspection - Continued.

Inspection	Method	Notes	Conditions	Symbol	Limits		Unit
					Min	Max	
Conformance inspection, part 2							
Heater current	1301	---		If	540	660	mA
Side terminal and base alignment	5101	---	Pin No. 3	---	---	---	---
Neck and bulb alignment (magnetic types)	5101	---		---	---	---	---
Face tilt	5101	---		---	---	---	---
Stray light emission	5216	---	Eb = 11,000 V dc; Ec2 = 770 V dc	---	---	---	---
Screens	5221	5/		---	---	---	---
Line width A (magnetic deflection)	5226	---	Ib = 200 µA dc	Width	---	0.30	mm
Line width C (magnetic deflection)	5226	---	Ib = 200 µA dc	Width	---	0.44	mm
Astigmatism	---	7/		Ratio	---	2	---
Focusing ampere turns	5246	---	Ib = 200 µA dc; D = 3.25 inches (82.55 mm)	AT	450	570	---
Heater-cathode leakage current	5251	---		---	---	---	---
Grid No. 2 leakage current	5251	---		---	---	---	---
Secureness of base, cap, or insert	1101	---		---	---	---	---
Base pin solder depth	1111	---		---	---	---	---
Permanence of marking	1105	---		---	---	---	---
Conformance inspection, part 3							
Life-test	---	---	Group C; Eb = 11,000 V dc; Ec2 = 770 V dc; Ib = 60 µA dc; t = 500 hours (min)	---	---	---	---
Life-test end points:	---						
Line width A	5226	---		Width	---	0.38	mm
Line width C	5226	---		Width	---	0.55	mm
Modulation	5223	---		ΔEc1	---	38	V dc

See footnotes at top of next page.

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TABLE I. Testing and inspection - Continued.

1/ When Ec2 is greater than 330 V dc or Ec1 is near zero, or both, the effective resistance of the anode and grid No. 3 supply should be adequate to limit the anode and grid No. 3 input power to 6 watts.

2/ The distance between the center of the unfocused, undeflected spot low intensity (Ec1 near cutoff) and the center of the image of the masking aperture observed at high intensity of the unfocused, undeflected spot shall not exceed the limit specified herein. Ec1 should not be held at zero for more than approximately 10 seconds to prevent damage to the screen.

3/ The test is made with maximum voltages applied to all electrodes connected through the base. Connections should be made in a manner that does not degrade the tubes electrical voltage breakdown characteristics. Criteria for acceptance is the absence of arc-over or corona.

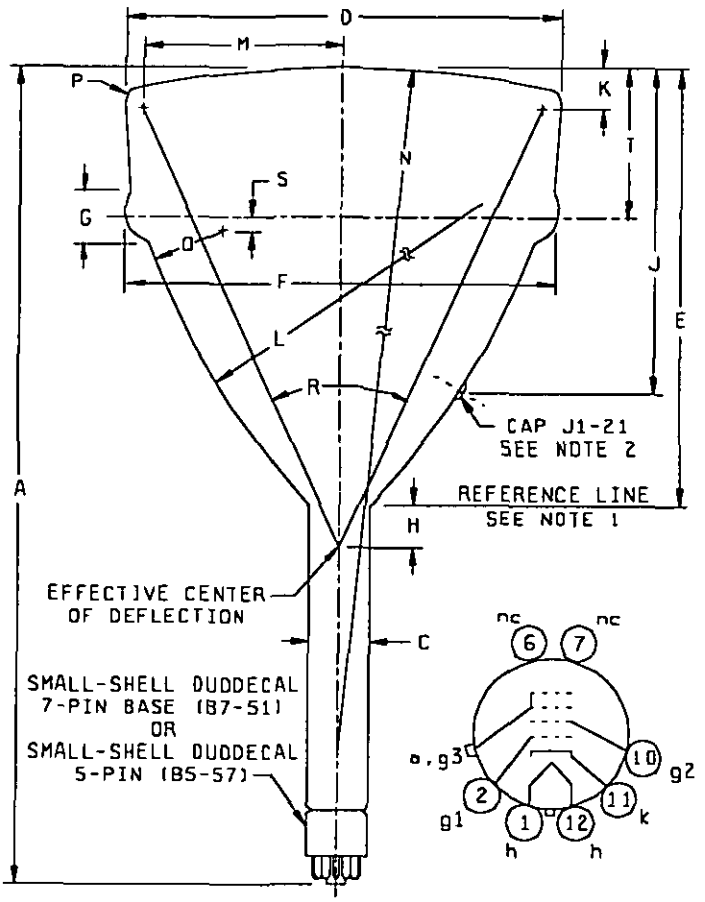
4/ Neck and base straightness shall be determined by the insertion of the tube neck in a cylinder 5 inches (127.0 mm) long and 1.500 +.003, -.000 inch (38.1 +0.08, -0.00 mm) inside diameter. This cylinder should move freely between the reference line and the base of the assembled tube.

5/ The screen characteristics shall be measured with constant beam energy of 0.24 watt defocused to a spot approximately 0.25 cm in diameter. The test conditions shall be anode voltage (relative to cathode) 400 V minimum and beam current 60 µA maximum. The screen characteristics shall comply with the following minimum limits: cb5 = 400 cb; G5: 1 = 4.

6/ This test to be performed at the conclusion of the holding period.

7/ With a single line of 1/60 second duration, and length 9 inches (228.6 mm), modulate grid No. 1 with 1 µs pulses. Adjust peak beam current to 200 µA and focus for the most circular spot at center of screen. Determine orientation for maximum and minimum spot diameters by rotating a partially merged raster, or by an equivalent method. Maintaining focus, determine line widths at these orientations in accordance with procedures described in method 5226. Astigmatism (the ratio of maximum to minimum line width) shall not exceed the value specified.

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Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
Conformance inspection, part 2				
A	17.630	18.000	447.80	457.20
C	1.380	1.500	35.05	38.10
D	10.380	10.620	263.65	269.75
E	9.250	9.620	234.95	244.35
F	---	10.620	---	269.75
G	---	.750	---	19.05
H	---	.950	---	24.13
J	6.870	7.620	174.50	193.55
Reference dimensions (see note 3)				
K	.650		16.51	
L	20.000 RAD		508.00 RAD	
M	4.500 RAD MIN useful screen		114.30 RAD MIN useful screen	
N	42.000 RAD		1066.80 RAD	
P	.380		9.65	
Q	2.00 RAD		50.80 RAD	
R	50°			
S	.242		6.15	
T	2.900		73.66	

NOTES:
1. Reference line determined by position where JEDEC gauge No. 112 will rest on bulb cone.
2. The plane through the tube axis and vacant pin position No. 3 may vary from the plane through the tube axis and anode terminal by an angular tolerance (measured about the tube axis) of 10°. Anode terminal is on the same side as vacant pin position No. 3.
3. These dimensions are for information only and are not required for inspection purposes.

FIGURE 1. Outline drawing of electron tube type 10AKP7.

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Navy - AS, CG, MC, OS

Preparing activity:
DLA - CC

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